

Vasileios Vlachakis



Address: Filopoimenos 56, 26221, Patra, Greece
Mob: +30 6976 245450
Tel/fax: +30 2615 502525
Email: vvlach@gmail.com
Date of Birth: Dec. 11, 1978
Nationality: Greek
Family: Married with two children

EDUCATION **2002:** Graduate of Department of **Civil Engineering**, University of Patras, Greece (Direction of Specialization: **Geotechnical and Hydraulic Engineering**).
2004: **Master of Science** in "Work of Infrastructure of Civil Engineer" of Postgraduate Studies Program, Department of Civil Engineering, University of Patras. Field of Study: Geotechnical Engineering.

MILITARY SERVICE **2005:** Hellenic Army - Engineers (November 2004 - August 2005).

WORK EXPERIENCE

11/2007 - today: **Co-founder and co-owner** of Geotechnical Engineering company: acquisition, management and planning of geotechnical and hydraulic projects for private and public sector in Greece.

5/2018 - 12/2025: **Head** of Design Planning & Works Department of Water Supply at Municipal Water Supply and Sewerage Company of Patras, Greece.

1/2008 - 7/2008: **Expert member** of applied research project of the University of Patras entitled: "Geological and Geotechnical Investigations in fire-ravaged municipalities of Ilia on the Recognition and Treatment of landslides - Design Protection Measures", with Scientific Coordinators Professors Mr. G. Koukis and G. A. Athanasopoulos.

3/2006 - 10/2007: **Civil Engineer** at the company: "TRITON Consulting Engineers Ltd." in Athens at the Geotechnical Engineering Division: Infrastructure Works (marine/ports, highways) and foundation building projects.

9/2002 - 9/2004: **Research assistant** at University of Patras, Dept. of Civil Engineering, Geotechnical Engineering Lab.: Participation in various applied research programs and working groups.

Areas of Interest and experience:

- Implementation of Laboratory Tests on Geotextiles (Thickness Measurements, Wide Width Tensile Strength, Trapezoidal Tear Strength, Grab Tensile test, Drop Cone test, Puncture test CBR-8mm, Burst Strength test).

	• Implementation of Soil Mechanics Laboratory Tests. Soil Classification tests, Atterberg Limits, Permeability tests, One-Dimensional Consolidation tests, Shear Strength tests: Direct Shear test, Triaxial tests (QU, UU, CUPP) and Special tests for the Determination of K_0 (CK_0 test). Rock Mechanics Laboratory Tests: QU, Triaxial tests, Point Load test. • Field Monitoring of Soil Behaviour with Measurements of Piezometers, Inclinometers, Tiltmeters, Strain-gages (Marine Works-Consolidation, Slope Stability, Structural). • Field monitoring of depth, temperature and conductivity of water into exploratory boreholes and wells. • Supervision of Sampling - Inquiring Drillings. • In-Situ tests (SPT- CPT). • Determination of Seismic Ground Velocity with Spectral Analysis of Surface Waves (SASW) Method. Site Measurements and Digital Processing of Measurements Results. • Technical Reports - Geotechnical Investigations - Design of Foundation Systems, Retaining Structures, Slopes and Embankments. • Numerical Analyses using the Finite Element Code (<i>PLAXIS</i>) for: Foundation Analyses (Determination of Soil Bearing Capacity and Settlements)- Slope Stability, Slope Reinforcement - Road Construction (<i>Cut and Cover - Top Down</i>), Deep Excavations, Sheet-Pile Walls, Marine Works, Seismic Analyses. • Water Distribution Analysis and Design of Water Supply Networks, using purpose special software as WaterGems.
TEACHING EXPERIENCE	• Laboratory Exercises in "Soil Mechanics" course of the 3rd year of Undergraduate Studies of Civil Engineering Department, University of Patras, 2002-2004.
SCHOLARSHIPS	2005: Academic Performance Scholarship for the Postgraduate Program of Department of Civil Engineering, University of Patras, from the State Scholarships Foundation of Greece (<i>IKY</i>).
PROFESSIONAL AFFILIATIONS	2003: Member of the Technical Chamber of Greece (<i>N.R.</i>: 93549). Member of the Association of Civil Engineers in Greece. 2008: Degree in Public Works Design, Ministry of Infrastructure & Transport (<i>N.R.</i>: 20384).
PRESENTATIONS - PAPERS	1. Tsitos, A.C., Vlachakis, V.S., Pelekis, P.C., Athanasopoulos, G.A., (2004), "Numerical Simulation of the Spectral Analysis of Surface Waves (SASW) Method – Parametric Analyses", <i>Proceedings of the 11th International Conference on Soil Dynamics & Earthquake Engineering (11th ICSDEE) and 3rd International Conference on Earthquake Geotechnical Engineering (3rd ICEGE)</i>, University of California, January 2004, Vol. 1, pp. 538-545, Stallion Press, Berkeley, USA. 2. Vlachakis, V.S., Athanasopoulos, G.A., (2005), "Application of Spectral Analysis of Surface Waves (SASW) Method in Slope Ground – Numerical Simulation and Parametric Analyses", 5th Hellenic Conference on Geotechnical and Geoenvironmental Engineering, Xanthi, Greece, May-June 2006. 3. Vlachakis V. S. and Panagopoulos N., (2006), "Analysis of a jetty formed by two rows of intersecting concrete piles", Presentation of a PLAXIS application at a

- Project (of TRITON Consulting Engineers S.A.), European Plaxis Users Meeting, Karlsruhe, November 2006, Germany.
4. Margaritis, B., Papaioannou, C., Theodoulidis, N., Savvaidis, A., Klimis, N., Makra, K., Demosthenus, M., Karakostas, M., Lekidis, V., Makarios, T., Salonikios, T., Athanasopoulos, G., Mylonakis, G., Papantonopoulos, C., Efthymiadou, V., Kloukinas, P., Ordonez, I., Vlachakis, V., Stewart, J., (2008), "Preliminary Report on the Principal Seismological and Engineering Aspects of the $M_w=6.5$ Achaia-Ilia (Greece) Earthquake on 8 June 2008", posted on the internet, http://www.geerassociation.org/GEER_Post%20EQ%20Reports/Greece_2008/Cover_Greece2008.html.
 5. Di Capua, G., Kayen, R., Kieffer, S., Button, E., Biscontin, G., Scasserra, G., Lanzo, G., Tommasi, P., Pagliaroli, A., Silvestri, F., d'Onofrio, A., Violante, C., Simonelli, A., Puglia, R., Mylonakis, G., Athanasopoulos, G., Vlachakis, V., Stewart, J., (2008), "Preliminary Report on the Seismological and Geotechnical Aspects of the April 6 2009 L' Aquila Earthquake in Central Italy", posted on the internet, http://www.geerassociation.org/GEER_Post%20EQ%20Reports/Italy_2009/Cover_Italy2009_Rev.html.
 6. Lanzo, G., Di Capua, G., Kayen, R.E., Kieffer, D.S., Button, E., Biscontin, G., Scasserra, G., Tommasi, P., Pagliaroli, A., Silvestri, F., d'Onofrio, A., Violante, C., Simonelli, A. L., Puglia, R., Mylonakis, G., Athanasopoulos, G., Vlachakis, V., Stewart, J. P., (2010), "Seismological and geotechnical aspects of the $M_w=6.3$ l' Aquila earthquake in central Italy on 6 April 2009", International Journal of Geoengineering Case histories, vol. 1, Issue 4, pp. 206-339.
 7. Vlachakis, V.S., Athanasopoulos, G.A., Pelekis, P.C., Spiliotopoulos, G., (2010), "Installation and Performance of a Steel Sheet Pile Wall for Supporting an Excavation in Urban Environment", Proceedings (in Greek) of 6th Hellenic Conference on Geotechnical and Geoenviromental Engineering, TEE - EEEEEFM, Volos, 2010.
 8. Pelekis, P.C., Vlachakis, V.S., Athanasopoulos, G.A., (2010), "Estimating the Size of a Ground Void Underneath a Building by the Surface Wave Method", Proceedings (in Greek) of 6th Hellenic Conference on Geotechnical and Geoenviromental Engineering, TEE - EEEEEFM, Volos, 2010.
 9. Athanasopoulos, G.A., Vlachakis, V.S., Zisimatou, E.P., (2010), "Numerical Analisis of Seismic Behavior of Free to Move Earth Retaining Walls Isolated with EPS Geofoam", Proceedings (in Greek) of 6th Hellenic Conference on Geotechnical and Geoenviromental Engineering, TEE - EEEEEFM, Volos, 2010.
 10. Batilas, A.V., Athanasopoulos, G.A., Pelekis, P.C., Vlachakis, V.S., Klimis, N.S., Mylonakis, G.E., (2010), "Soil Liquefaction at the Coastal Zone of Kato Achaia in the Achaia-Ilia, Greece, Earthquake of June 8, 2008", Proceedings (in Greek) of 6th Hellenic Conference on Geotechnical and Geoenviromental Engineering, TEE - EEEEEFM, Volos, 2010.
 11. Athanasopoulos, G.A., Margaritis, B., Klimis, N., Mylonakis, G., Theodoulidis, N., Papaioannou, C., Savvaidis, A., Efthimiadou, V., Vlachakis, V.S., Papandonopoulos, C., Stewart, J.P., (2010), "Achaia-Ilia ($M_w=6.4$) Earthquake of June, 8, 2008: Reconnaissance & Geotechnical Aspects", Proceedings (in Greek)

- of 6th Hellenic Conference on Geotechnical and Geoenviromental Engineering, TEE - EEEEFM, Volos, 2010.
12. Athanasopoulos, G.A., Vlachakis, V.S., Pelekis, P.C., (2011), "Installation and Performance of a Steel Sheet Pile Wall for Supporting an Excavation in Urban Environment", Geo-Frontiers 2011, Dallas, Texas, March, 2011.
 13. Athanasopoulos-Zekkos, A., Vlachakis, V.S., and Athanasopoulos, G.A. (2013), "Phasing issues in the seismic response of yielding, gravity-type earth retaining walls – Overview and results from a FEM study", Soil Dynamics and Earthquake Engineering 55 (2013) 59-70.
 14. Athanasopoulos, G., Vlachakis, V., Zekkos, D. and Spiliotopoulos, G., (2013), "The December 29th 2010 Xerolakka Municipal Solid Waste Landfill Failure", Proceedings of the 18th International Conference on Soil Mechanics and Geotechnical Engineering, Paris 2013.
 15. Theodoulidis, N., Savvaidis, A., Margaritis, B., Zargli, E., Klimis, N., Lazaridis, T., Stewart, J.P., Athanasopoulos, G., Pelekis, P., Vlachakis, V., and Batilas, A., (2013), "Geologic and Terrain-Based Proxy Estimation of VS30 in Greece", Proceedings of the Joint Assembly IAHS, IAPSO, IASPEI, Knowledge for the Future, Gothenburg, Sweden, 22-26 July, 2013.
 16. Zekkos, D., Vlachakis, V. S., and Athanasopoulos, G. A. (2014). "The 2010 Xerolakka Landfill Slope Instability", Environmental Geotechnics, Vol 1, Issue EG1, pp.56-65, ICE Publishing.
 17. Batilas, A., Pelekis, P., Vlachakis, V., and Athanasopoulos, G., (2014), "Soil Liquefaction/Nonliquefaction in the Achaia-Ilia (Greece) 2008 Earthquake: Field Evidence, Site Characterization and Ground Motion Assessment", International Journal of Geoengineering Case histories, <http://casehistories.geoengineer.org>, Vol.2, Issue 4, p.270-287, doi: 10.4417/IJGCH-02-04-03.
 18. Vlachakis, V.S., Athanasopoulos, G.A., and Athanasopoulos-Zekkos, A. (2014), "Seismic Displacements of yielding gravity-type earth retaining walls-Results from a FEM Study", ASCE, GSP No 234, pp. 3188-3197.
 19. Nikolaou S., Zekkos, D., Assimaki, D. and Gilsanz R. (editors) and 84 Contributing Authors (2014), "GEER/EERI/ATC Earthquake Reconnaissance, January 26th/February 2nd 2014 Cephalonia, Greece Events", Version 1: June 6, 2014, posted on the internet: http://www.geerassociation.org/GEER_Post%20EQ%20Reports/Cephalonia_Greece_2014/index.html.
 20. Kitsis, V.G., Vlachackis, V.S., Athanasopoulos, G.A., Athanasopoulos – Zekkos, A., (2015), "Seismic Thrust vs Wall Inertia in Non – Yielding Retaining Walls Under Earthquake Loading: Synchronous or Asynchronous Action?", Proceedings of IFCEE 2015, San Antonio, Texas, USA, March 17-21, 2015.
 21. Vlachakis, V.S., Pelekis, P.C., Batilas, A.V. and Athanasopoulos, G.A. (2015), "Shear Wave Velocity Measurements in The Area of University of Patras Accelerograph Network (UPAN) Using Active & Passive Surface Wave Methods", Proceedings of SAGEEP 2015, Environmental and Engineering Geophysical Society, Austin, Texas, USA, March 22-26, 2015.

SELECTED STUDIES

1. New Port of Patras (Part-2), "Evaluation of Geotechnical Conditions in the Area of Quay Walls Following the Pre-Loading of the Sea Bottom – Stability Analyses", Technical Report to Parnon S.A. September, 2003.
2. Athanasopoulos, G., A., and Pelekis, P. C., (2003), "Experimental Evaluation of Friction Angle of the Breakwater Quarry Run – Stability Analyses", New Port of Patras (2nd Section), Technical Report submitted to Parnon S.A., Dec. 2003.
3. Athanasopoulos, G.A, Vlachakis, V.S., Xenaki, V.C., (2004), "Experimental Investigation of Local Soil Formation Behavior with Special Tests (K0 Triaxial Loading) for the Extension of Athens Metro", Submitted to Edafomichaniki S.A., Athens, November 2004.
4. Port of Fujairah. South Breakwater Berths (SBB). Design and Slope Stability Report for Filter Layers Behind Existing Breakwater. Technical Report for ATHENA S.A., November 2006.
5. Aeolian Park at Mt. Pantokratoras, Corfu, Greece. Field Investigation for the Detection of Underground Cavities. Technical Report to G. Armenis, August 2008.
6. Athanasopoulos, G.A. and Vlachakis, V.S., (2008), "Experimental Investigation of Local Soil Formation Behavior with Special Laboratory Tests for the Thessaloniki Metro", Submitted to J/V AEGEK - Impregilo - Seli, Thessaloniki, April 2008.
7. Analyses and Monitoring of Deep Supported Excavation for the Project: "Argyris Market", Technical Report to Triaina S.A., March, 2008.
8. Patras Ring Road, "Evaluation of Geotechnical Investigations Results - Slope Stability Analyses - Study of Proposed Soil Retaining Structures KM 1+716 to 1+940, Technical Report to the Regional Operational Program Special Service, Region of Western Greece, May 2008.
9. Stavros Niarchos Foundation Cultural Center at the Athens Faliron Delta, Greece, "Laboratory Evaluation of Liquefaction Resistance, Strength and Dynamic Moduli/Damping Ratios of Soil Formations Underlying the Project Site", Technical Report to Stavros Niarchos Foundation, July 2010.
10. "Crosshole Measurements at two Locations of the Project Site" for the Project: Ptolemais Unit V 660MW Power Plant, Greece, Technical Report to OMETE EDAFOSTATIKI S.A., November 2013.
11. "Evaluation Of Vs-Depth Profiles at the Project Site by SASW Measurements", Technical Report to J&P AVAX S.A., September 2015.
12. Trans Adriatic Pipeline Project (TAP), "AC03 Seismic Site Measurements – Factual and Interpretative Report", Technical Report to J/V Renco-Terna, July 2016.
13. Trans Adriatic Pipeline Project (TAP), "AC02 Seismic Site Measurements – Factual and Interpretative Report", Technical Report to J/V Renco-Terna, July 2016.
14. "Geotechnical Study: Retaining Wall of Pantokratoros Slope in front of 3rd Primary School of Patras", Geotechnical Study to Patras Municipality, July, 2016.
15. "Foundation Design and Analysis of Building S1, at 5* Hotel at Ag. Sofia, Mytikas, Aitolokarnanias", Geotechnical Study to Aithon S.A., Aug. 2016.
16. Nigeria AFAM [8] TM2500+ (Federal Ministry of Power of Nigeria), "Measurement of Ground Vibrations Induced by Deep Dynamic Compaction (Area C)", Technical Report to METKA S.A., February 2017.

17. "Rockfall Protection Design of Regional Road No.5 at the Stromi Area", Geotechnical Study to the Region of Central Greece, December 2017.
18. "Rockfall Protection Design of Area 11 at Regional Road of Xarakas – Assos, Cefalonia, Greece", Geotechnical Study to the Region of Ionian Islands, Greece, April 2019.
19. "Landslide Rehabilitation Study at National road n. 33 Patra-Tripotama-Junction with N.R.74, at responsibility area of the Region of Western Greece", Geotechnical Study to Region of Western Greece, October, 2020.
20. "Landslide Rehabilitation Study at National road n. 33 Patra-Tripotama-Junction with N.R.74, at responsibility area of the Region of Western Greece", Structural Study to Region of Western Greece, October, 2020.
21. "Geotechnical Investigation and Foundation Conditions Assessment Using Geophysical Methods for the New Building ST of Hellenic Open University at Perivola, Patra", Geotechnical Study to Hellenic Open University, Patras, January 2021.
22. "Geotechnical Investigation & Study, Rainwater Management in the Area of Roman Stadium at Patras", Geotechnical Study for Ephorate of Antiquities of Achaia, Patras, February 2021.
23. "Soil Retaining System Study for the Construction of Building E' of Hellenic Open University, at Perivola, Patra", Geotechnical Study for Hellenic Open University, Patras, March 2023.
24. "Update of Landslide Rehabilitation Study at Limenia area of Poros Port, Cefalonia", Geotechnical Study to Region of Ionian Islands, March 2024.
25. "Underwater Autopsy of Lighthouse Pier at Aigion Port", Technical Report to Civil Engineer Makis Kolokythas, May 2024.
26. "Reinforcement/Rehabilitation of Retaining Wall of 1st Cemetery of Patras Municipality", Geotechnical Study to Anodos ATE, Patras July 2025.

FOREIGN LANGUAGES

- **ENGLISH:** 1993: Cambridge First Certificate in English.
- **FRENCH:** 1994: Certificat de Langue Française (Delf A1, A2, A3, A4).

COMPUTER KNOWLEDGE

- Windows, Word, Excel, Power Point, Origin, Fortran 77, AutoCad, Photoshop.
- Geotechnical Finite Element Code of Analysis, *PLAXIS* v.9.
- Geotechnical Engineering Analysis program LARIX-5.
- Rockfall Protection Design Software Rocscience - RocFall v.6
- Water Distribution Analysis and Design Software Bentley - WaterGems v.10

OTHER SKILLS - INTERESTS

- 2017:** Scuba Diving courses and acquisition of Advanced Open Water Diver of ANDI International.
- 2012:** Rock Climbing courses and acquisition of pertinent degree.
- 1999:** Black & White Photography courses in the Photographic Section Cultural of Students at Patras University and attendance in photographic exhibitions (1999-2002).
- 1998:** Scuba Diving courses and acquisition of Degree of Scuba Diver 1* of CMAS.

January 2026